Monad Mini PC Node System Report

Generated: 2026-05-30 06:25:31 MDT
Privacy posture: public report. Hostnames, usernames, IP addresses, keys, signatures, wallet-like hex strings, and token-like values are redacted or omitted.

Executive Summary

Monad services: monad-bft=active, monad-execution=active, monad-rpc=active.
Estimated Monad service CPU over the prior interval: approximately 615.6% of one CPU core.
Thermals: max 91.1 C, average 43.6 C.
PDF resource-window screenshot: included (captured window: System Monitor).

System Specs

CPU: AMD Ryzen 7 8745H w/ Radeon 780M Graphics
CPU count reported by kernel: 16
Online CPU range: 0-7
RAM: 44Gi
Root disk: 915G total, 203G used, 666G available, 24% used
Kernel: 6.17.0-1023-oem

Monad Node Runtime

Services:
- monad-bft: active=active, state=running, main_pid=34003, restarts=0, memory=1150.7 MB, cpu_since_start=27942.8 sec
- monad-execution: active=active, state=running, main_pid=33969, restarts=0, memory=25310.7 MB, cpu_since_start=3857193.7 sec
- monad-rpc: active=active, state=running, main_pid=33959, restarts=0, memory=585.9 MB, cpu_since_start=87010.4 sec

Processes: no visible Monad process rows were returned by ps.

Temperature And Fluctuation Analysis

- Current thermal snapshot: average 43.6 C, maximum 91.1 C across 11 reported sensors.
- Four-hour comparison: 2.5 C increase in max temperature from the prior report.
- Thermal fluctuation note: change stayed within a small operating band for this sample window.
- 24-hour observed max-temperature range: 88.5 C to 91.1 C.

Top reported sensor readings:
- k10temp-pci-00c3 / Tctl: 91.1 C
- amdgpu-pci-c500 / edge: 59.0 C
- nvme-pci-0100 / temp1: 59.0 C
- spd5118-i2c-0-50 / temp1: 43.5 C
- spd5118-i2c-0-51 / temp1: 37.8 C
- nvme-pci-0400 / Sensor 2: 36.9 C
- nvme-pci-0100 / Composite: 33.9 C
- nvme-pci-0100 / Sensor 1: 33.9 C
- nvme-pci-0400 / Sensor 1: 32.9 C
- nvme-pci-0400 / Composite: 31.9 C
- acpitz-acpi-0 / temp1: 20.0 C

System And Node Events Cross-Check

Recent sanitized events from the last four hours, classified for public context:
- [Report collection issue] monad-bft: local report probes could not collect a non-critical data point.

  Sanitized detail: probe failed: Command '['journalctl', '-u', 'monad-bft', '--since', '4 hours ago', '--no-pager']' timed out after 20 seconds
- [Report collection issue] monad-execution: local report probes could not collect a non-critical data point.
  Sanitized detail: probe failed: Command '['journalctl', '-u', 'monad-execution', '--since', '4 hours ago', '--no-pager']' timed out after 20 seconds

Security Change Log

Recent operator security changes, sanitized for public sharing:
- 2026-05-07: Restored Monad runtime settings with systemd drop-ins instead of editing vendor unit files.
  Why: Keeps the node configuration easier to audit and preserve across package updates while restoring the known-good event-ring, RPC websocket, and CPU-placement behavior from the earlier install.
  Privacy note: No command lines, passwords, keys, peer addresses, node identities, wallet-like values, or public IP addresses are recorded here.
- 2026-05-07: Restricted Monad secret-bearing files to owner-only read/write permissions.
  Why: The local keystore environment file and node identity key files were more broadly readable than necessary. Owner-only permissions reduce local accidental disclosure risk without changing node behavior.
  Privacy note: File paths are summarized by purpose; key contents and unlock values are never included.
- 2026-05-07: Confirmed normal local users cannot read the Monad BFT process command line or environment through procfs.
  Why: The BFT binary currently accepts its keystore unlock value through a command-line option, so process-argument visibility matters. Existing procfs hiding reduces exposure to non-privileged local users.
  Privacy note: The actual process command line and keystore unlock value are intentionally omitted.
- 2026-05-07: Documented the remaining upstream security limitation for Monad BFT keystore unlock handling.
  Why: A cleaner design would let operators pass the keystore unlock value through a protected file, stdin, systemd credential, or similar non-argv mechanism. The current local binary help only exposes a command-line unlock option.
  Privacy note: This notes the design limitation only; no secret [redacted] is recorded.
- 2026-05-07: Kept public report generation sanitized and avoided raw systemctl status or full process command-line capture.
  Why: Full status and process listings can expose sensitive command-line arguments. Reports use narrower systemd fields, sanitized journal excerpts, and explicit redaction before writing TXT/PDF outputs.
  Privacy note: Reports redact hostnames, usernames, IP addresses, keys, signatures, wallet-like hex strings, token-like values, and sensitive argument values.
- 2026-05-10: Added a local OpenTelemetry Collector for the Monad metrics endpoint.
  Why: Monad BFT requires a telemetry endpoint argument. The previous configuration pointed at a local endpoint with no listener, causing repeated connection-refused errors. A localhost-only collector now provides the required endpoint without exporting private telemetry externally.
  Privacy note: The report records only the operational change and reason. No raw telemetry payloads, peer identities, public IP addresses, keys, command lines, or unlock values are included.

Research Notes

- This report is generated from local system telemetry and Monad service logs on the mini PC.
- Values are snapshots, not lab-grade benchmarks. Trends become more useful after multiple four-hour samples.
- The report intentionally avoids private identity material, validator keys, wallet-like

values, raw peer lists, public IPs, and command lines containing sensitive arguments.
- Intended use: public community reporting about running a Monad node on smaller hardware while retaining useful operational data for technical readers.

System Resource Window Snapshot